

ASSIGNMENT – 10.5

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Batch - 03

Task 1:-

```
1  # Assignment 10.5: AI Assisted Coding - Code Review and Quality Improvement
2
3  # Task 10.1: Variable Naming Issues
4  # Original Code:
5  # def f(a, b):
6  #     return a + b
7  # print(f(10, 20))
8
9  # Improved Code with meaningful variable and function names
10 def add_numbers(first_number, second_number):
11     """
12     Adds two numbers and returns the result.
13
14     Args:
15         first_number (int or float): The first number to add.
16         second_number (int or float): The second number to add.
17
18     Returns:
19         int or float: The sum of the two numbers.
20     """
21     return first_number + second_number
22
23 # Example usage for Task 1
24 print("Task 1 Output:")
25 print(add_numbers(10, 20))
26 print()
27
28 # Task 10.2: Missing Error Handling
29 # Original Code:
30 # def divide(a, b):
31 #     return a / b
32 # print(divide(10, 0))
```

```

# Task 10.2: Missing Error Handling
# Original Code:
# def divide(a, b):
#     return a / b
# print(divide(10, 0))

# Improved Code with error handling
def divide_numbers(dividend, divisor):
    """
    Divides two numbers and returns the result.

    Args:
        dividend (int or float): The number to be divided.
        divisor (int or float): The number to divide by.

    Returns:
        float: The quotient of the division.

    Raises:
        ValueError: If divisor is zero.
        TypeError: If inputs are not numbers.
    """
    try:
        if not isinstance(dividend, (int, float)) or not isinstance(divisor, (int, float)):
            raise TypeError("Both dividend and divisor must be numbers.")
        if divisor == 0:
            raise ValueError("Cannot divide by zero.")
        return dividend / divisor
    except ZeroDivisionError:
        raise ValueError("Cannot divide by zero.")
    except TypeError as e:
        raise TypeError(f"Invalid input types: {e}")

```

Output:-

```

PS C:\Users\hp\OneDrive\Desktop\ai> & 'C:\Users\hp\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51943' '--' 'C:\Users\hp\OneDrive\Desktop\ai\10.py'
Task 1 Output:
30

```

Task 2:

```

print("Task 2 Output:")
try:
    print(divide_numbers(10, 2))
    print(divide_numbers(10, 0)) # This will raise an error
except ValueError as e:
    print(f"Error: {e}")
print()

# Task 10.3: Student Marks Processing System
# Original Code:
# marks=[78,85,90,66,88]
# t=0
# for i in marks:
#     t=t+1
# a=t/len(marks)
# if a>=90:
#     print("A")
# elif a>=75:
#     print("B")
# elif a>=60:
#     print("C")
# else:
#     print("F")

# Improved Code following PEP 8, with meaningful names, functions, comments, and error handling

def calculate_average(marks):
    """
    Calculates the average of a list of marks.

    Args:
        marks (list of int or float): List of student marks.

    Returns:
        float: The average of the marks.

    Raises:
        ValueError: If marks list is empty or contains non-numeric values.
    """
    if not marks:
        raise ValueError("Marks list cannot be empty.")
    if not all(isinstance(mark, (int, float)) for mark in marks):
        raise ValueError("All marks must be numbers.")

    total = sum(marks)
    average = total / len(marks)
    return average

def get_grade(average):
    """

```

```

def get_grade(average):
    """
    Determines the grade based on the average mark.

    Args:
        average (float): The average mark.

    Returns:
        str: The grade (A, B, C, or F).
    """
    if average >= 90:
        return "A"
    elif average >= 75:
        return "B"
    elif average >= 60:
        return "C"
    else:
        return "F"

def process_student_marks(marks):
    """
    Processes student marks to calculate total, average, and grade.

    Args:
        marks (list of int or float): List of student marks.

    Returns:
        dict: A dictionary containing total, average, and grade.
    """
    try:
        total = sum(marks)
        average = calculate_average(marks)
        grade = get_grade(average)
        return {
            "total": total,
            "average": average,
            "grade": grade
        }
    except ValueError as e:
        print(f"Error processing marks: {e}")
        return None

```

Output:-

```

Task 2 Output:
5.0
Error: Cannot divide by zero.

```

Task 3:-

```
def factorial(n):  
    """  
    Calculates the factorial of a non-negative integer.  
  
    The factorial of a number n is the product of all positive integers  
    less than or equal to n. For example, factorial(5) = 5 * 4 * 3 * 2 * 1 = 120.  
  
    Args:  
        n (int): A non-negative integer for which to calculate the factorial.  
  
    Returns:  
        int: The factorial of n.  
  
    Raises:  
        ValueError: If n is negative.  
        TypeError: If n is not an integer.  
    """  
    # Validate input  
    if not isinstance(n, int):  
        raise TypeError("Input must be an integer.")  
    if n < 0:  
        raise ValueError("Factorial is not defined for negative numbers.")  
  
    # Initialize result to 1 (factorial of 0 is 1)  
    result = 1  
  
    # Multiply result by each integer from 1 to n  
    for i in range(1, n + 1):  
        result *= i # Accumulate the product  
  
    return result
```

Output:

```
Task 3 Output:  
Total: 407  
Average: 81.40  
Grade: B
```

Task 4:-

Input:-

```
# Initialize feedback list
feedback = []

# Check minimum length
if len(password) < 8:
    feedback.append("Password must be at least 8 characters long.")
else:
    feedback.append("✓ Minimum length requirement met.")

# Check for uppercase letter
if not re.search(r'[A-Z]', password):
    feedback.append("Password must contain at least one uppercase letter.")
else:
    feedback.append("✓ Contains at least one uppercase letter.")

# Check for lowercase letter
if not re.search(r'[a-z]', password):
    feedback.append("Password must contain at least one lowercase letter.")
else:
    feedback.append("✓ Contains at least one lowercase letter.")

# Check for digit
if not re.search(r'\d', password):
    feedback.append("Password must contain at least one digit.")
else:
    feedback.append("✓ Contains at least one digit.")

# Check for special character
if not re.search(r'[!@#$%^&*(),.?":{}|<>]', password):
```

```
def main():
    """
    Main function to run the password validation program.
    """
    # Get password input from user
    password = input("Enter password: ")

    # Validate the password
    is_valid, feedback = validate_password(password)

    # Display result
    if is_valid:
        print("Password is strong!")
    else:
        print("Password is weak. Here's why:")

    print(feedback)
```

Output:-

```
Task 4 Output:  
120
```

Task 5:

```
# Example usage for Task 5 (non-interactive for demonstration)  
print("Task 5 Output:")  
# Test with a strong password  
test_password = "StrongPass123!"  
is_valid, feedback = validate_password(test_password)  
print(f"Testing password: {test_password}")  
✓ if is_valid:  
|     print("Password is strong!")  
✓ else:  
|     print("Password is weak. Here's why:")  
print(feedback)  
print()
```

Output:-

```
Task 5 Output:  
Testing password: StrongPass123!  
Password is strong!  
✓ Minimum length requirement met.  
✓ Contains at least one uppercase letter.  
✓ Contains at least one lowercase letter.  
✓ Contains at least one digit.  
✓ Contains at least one special character.
```